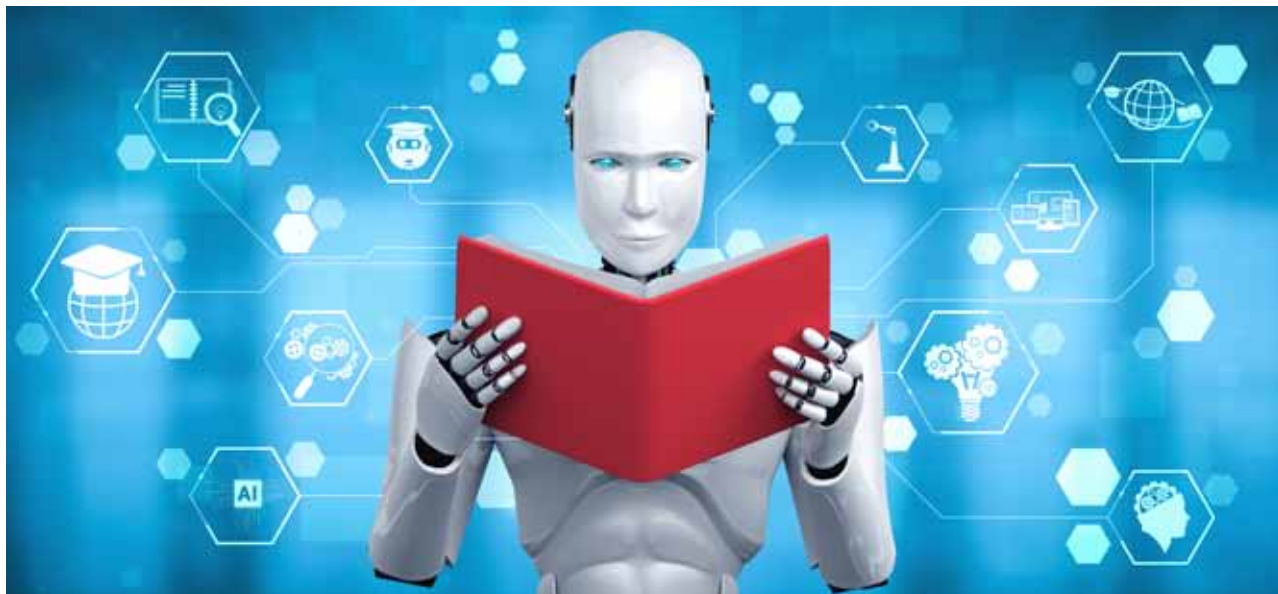


AI: An Introduction to Scikit-learn and Our First Trained Model



In the last article in this series on AI and machine learning, we started our discussion of neural networks by using TensorFlow. We also got familiar with our first data set by using Keras. In this seventh article in the AI series, we will continue exploring neural networks and the use of data sets for training models. We will also introduce yet another powerful Python library for machine learning called scikit-learn. But we will begin by discussing two sensational applications that will show us the power of AI and machine learning.

penAI is an artificial intelligence (AI) research laboratory which does a lot of research in the fields of AI and machine learning. Elon Musk is one of the founding members of this organisation. In November 2022, the lab launched a new online tool called ChatGPT. It is an AI chatbot that can chat like a human being. This chatbot is a large language model trained with both supervised and reinforcement learning paradigms of machine learning. ChatGPT uses OpenAI's GPT-3.5 language model, an improved version of GPT-3 (Generative Pre-trained Transformer 3), a language model that uses deep learning to produce human-like text. I still feel the excitement I felt while using ChatGPT for the first time. The application clearly demonstrates the power of AI. The answers given by the chatbot are of high

quality and often indistinguishable from a reply given by a human being. You can use ChatGPT to correct grammatical errors, rephrase sentences, summarise passages, write programs, etc. Indeed, I have rephrased many of the sentences in this article using ChatGPT. Further, I have tested ChatGPT using text with intentional grammar mistakes, not the usual ones I commit, and the corrected sentences were highly accurate. The ability to re-phrase and summarise passages was also phenomenal.

It might be even possible that programmers could solve programming puzzles within a very short time using ChatGPT. One such claim was made in a recent programming contest called 'Advent of Code 2022'. In fact, in December 2022, just after one month of ChatGPT's existence, Stack Overflow banned its use for generating answers.